

ESWIN

EBC77 Series Single Board Computer Datasheet

Rev 2.0

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Notice

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Change History

Version	Date	Description
Rev 2.0	2026/03/24	Restructured document layout
Rev 1.0	2025/04/25	Initial release

Contents

Notice **1**

Intellectual Property Statement 1

Limitation of Liability 1

Change History **2**

1. Product Overview **6**

2. Hardware Specifications **7**

3. Design Block Diagram **8**

3.1. EIC7700 Chip Block Diagram 8

3.2. System Block Diagram 9

3.3. Power Block Diagram 10

4. Interface and Connector Definitions **12**

4.1. 40-Pin IO Expansion Header 12

4.2. MIPI CSI Camera Interface (J10) 14

4.2.1. C-PHY Mode 16

4.2.2. D-PHY Mode 16

4.3. MIPI DSI/CSI Multiplexed Interface (J9) 17

4.4. PCIe Gen3 x4 Expansion Interface (J6) 19

4.5. Fan Interface (J2) 21

4.6. RTC Battery Interface (J12) 22

5. Mechanical Dimensions **24**

List of Figures

1	EIC7700 Chip Block Diagram	8
2	SBC System Block Diagram	9
3	SBC Power Tree	10
4	SBC Power-On Sequence	11
5	40-Pin IO Expansion Header	12
6	MIPI CSI Interface	14
7	C-PHY Connection Diagram	16
8	D-PHY Connection Diagram	16
9	MIPI DSI/CSI Interface Diagram	18
10	PCIe Interface Diagram	20
11	Fan Interface Diagram	22
12	RTC Battery Interface	23
13	Mechanical Dimensions	24

List of Tables

1	SBC Key Hardware Parameters	7
2	SBC 40-Pin Expansion Header Pin Definitions	12
3	EIC7700 C-PHY / D-PHY Mode Description	14
4	EIC7700 D-PHY 2/4 Lane Mode	17
5	MIPI DSI Interface Pin Definitions	19
6	PCIe Interface Pin Definitions	21

1. Product Overview

The EBC77 series single-board computer (hereinafter referred to as SBC) is a high-performance, highly integrated computing platform designed for edge AI computing, multimedia processing, and compact embedded applications. It is built on ESWIN's latest EIC7700 AI SoC, which integrates high-performance SiFive RISC-V CPU cores and a powerful AI acceleration unit.

The SBC adopts an innovative high-density, single-board design, integrating a complete system—including CPU, memory, storage, rich multimedia and high-speed interfaces—on an ultra-compact board measuring just 85mm x 56mm. This design eliminates modular separation, achieving extreme compactness, making it the ideal “heart” for building space-constrained intelligent terminals, edge boxes, AI cameras, and industrial control equipment.

With its flagship-grade hardware configuration, ultra-compact form factor, and comprehensive interface support, the SBC can be seamlessly integrated into various end-products with strict space and form factor requirements, providing a powerful and reliable hardware foundation from prototype validation to mass deployment.

2. Hardware Specifications

The SBC integrates flagship-grade hardware in an ultra-compact form factor, designed to deliver powerful hardware support for edge AI inference, multimedia processing, and high-performance embedded computing.

Table 1: SBC Key Hardware Parameters

Parameter	Description
SOC	ESWIN EIC7700
CPU	64-bit OoO RISC-V RV64GC Quad-Core @ 1.4GHz, up to 1.8GHz
AI Processor	Up to 13.3 TOPS, supporting INT8/INT16/FP16
Video Processing	Supports HEVC (H.265) and AVC (H.264) hardware encoding/decoding: H.265 Decode: 8K@56fps or 36-ch 1080p@25fps H.265 Encode: 8K@25fps or 16-ch 1080p@25fps
Memory	Onboard 8GB/16GB/32GB LPDDR5 @ 6400Mbps
Storage	1x 8MB SPI Flash
	1x Micro SD Card Slot
Networking	1x Gigabit Ethernet RJ45 Port
	1x Onboard 802.11ac Dual-Band Wi-Fi Module
Display	1x Micro HDMI 2.0 Output
	1x 4-lane MIPI DSI TX (Compatible with MIPI CSI RX)
Camera	1x 4-lane MIPI CSI RX (Compatible with MIPI DSI TX)
	1x 4-lane MIPI CSI RX
Audio	Via 40-pin Expansion Header
PCIe	1x 4-lane PCIe Gen 3.0 FPC Connector
USB	2x USB 3.0 Type-A Ports (Stacked)
	2x USB 2.0 Type-A Ports (Stacked)
RTC	1.0mm 2-pin 3V Connector
Fan	1.0mm 4-pin 5V Connector
Buttons	1x Reset Button
	1x Boot Mode Select Button
Power Supply	USB Type-C PD, Recommended $\geq 45W$ (15V@3A)
Debug	1x Micro-B Debug Port
Other I/O	I ² C, I ² S, SPI, UART, GPIO (Mapped to 40-pin Expansion Header)
Dimensions	85mm x 56mm

Tip

For more information about the EIC7700 SoC:
The CPU, NPU, video encoding/decoding, and other computing and processing capabilities described in this document are all provided by the onboard ESWIN EIC7700 AI SoC. For the complete architecture, register definitions, and low-level technical details of this SoC, please refer to the standalone EIC7700 Chip Datasheet.

3. Design Block Diagram

3.1. EIC7700 Chip Block Diagram

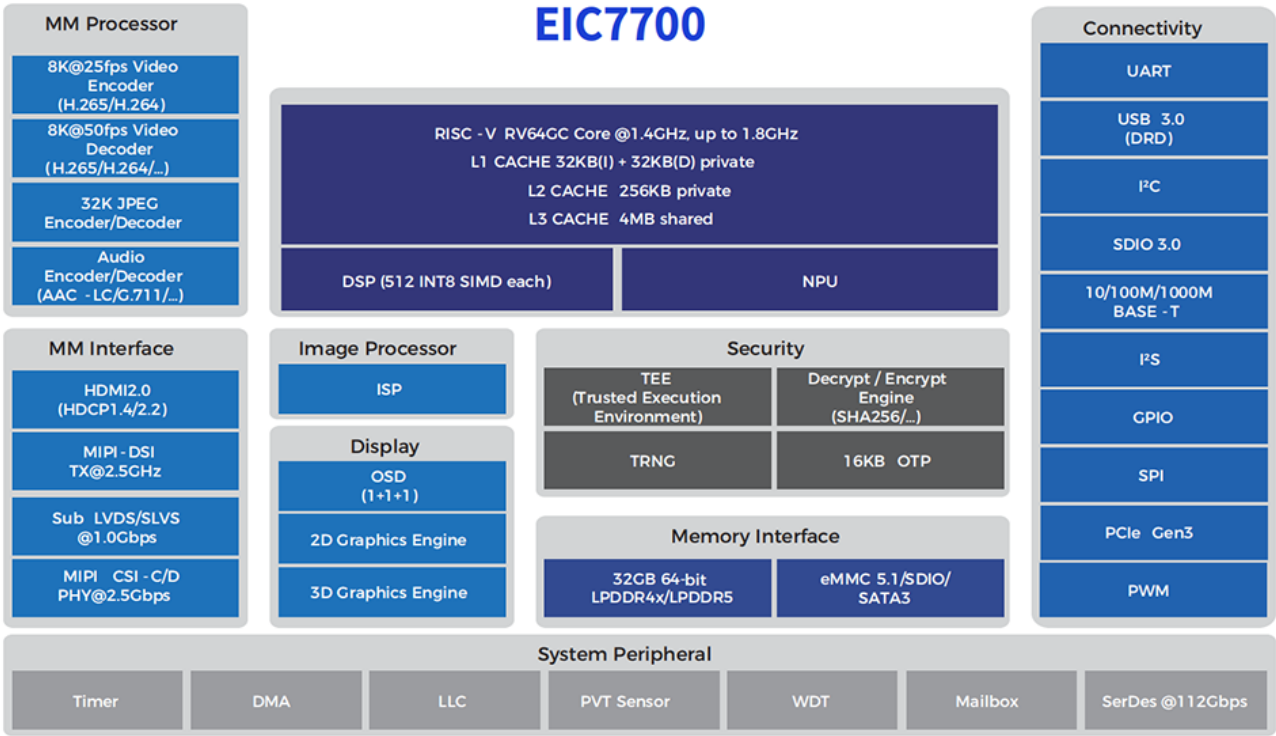


Figure 1: EIC7700 Chip Block Diagram

3.2. System Block Diagram

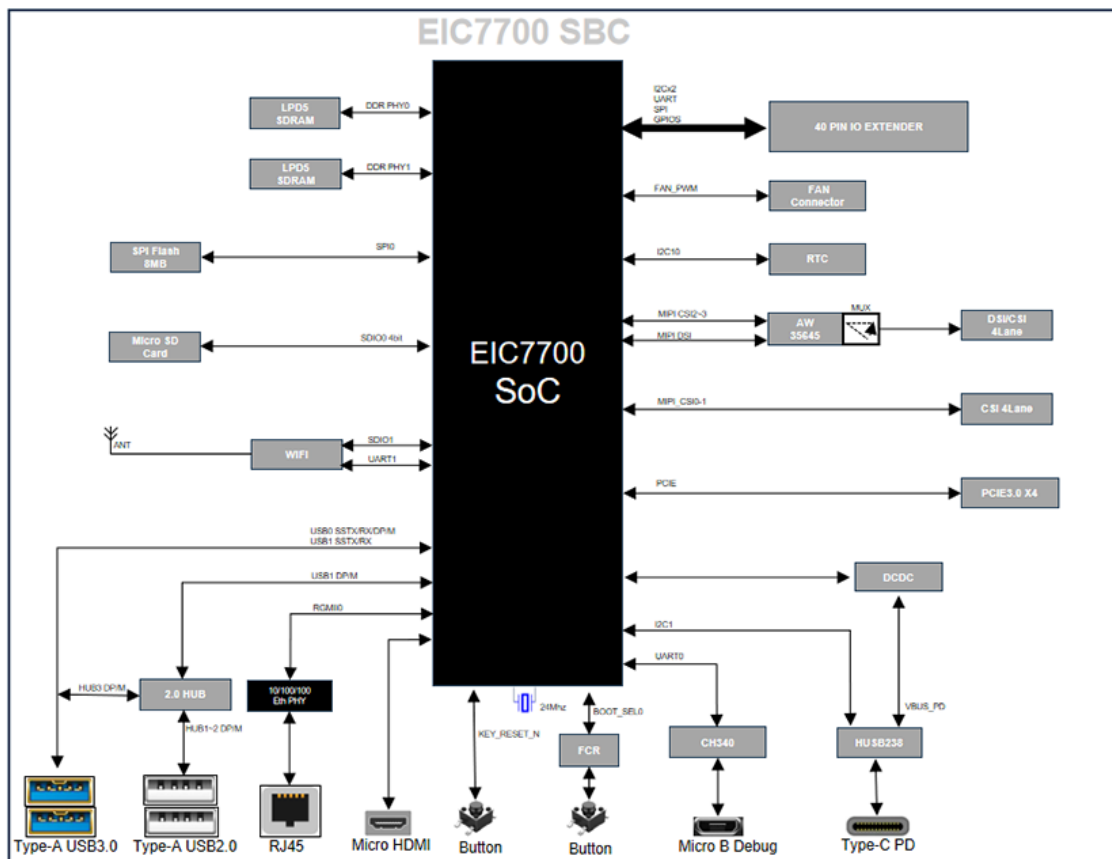


Figure 2: SBC System Block Diagram

3.3. Power Block Diagram

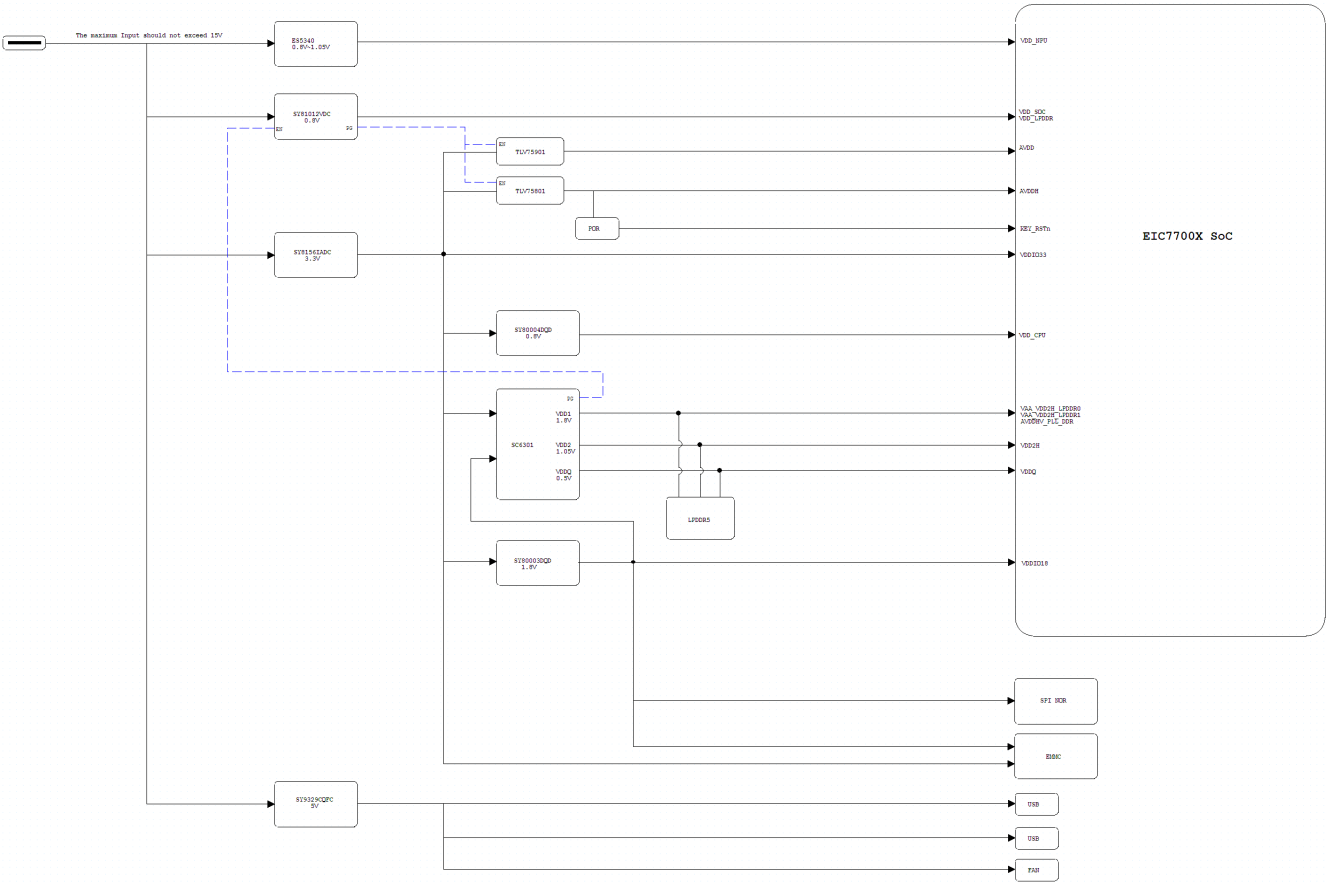


Figure 3: SBC Power Tree

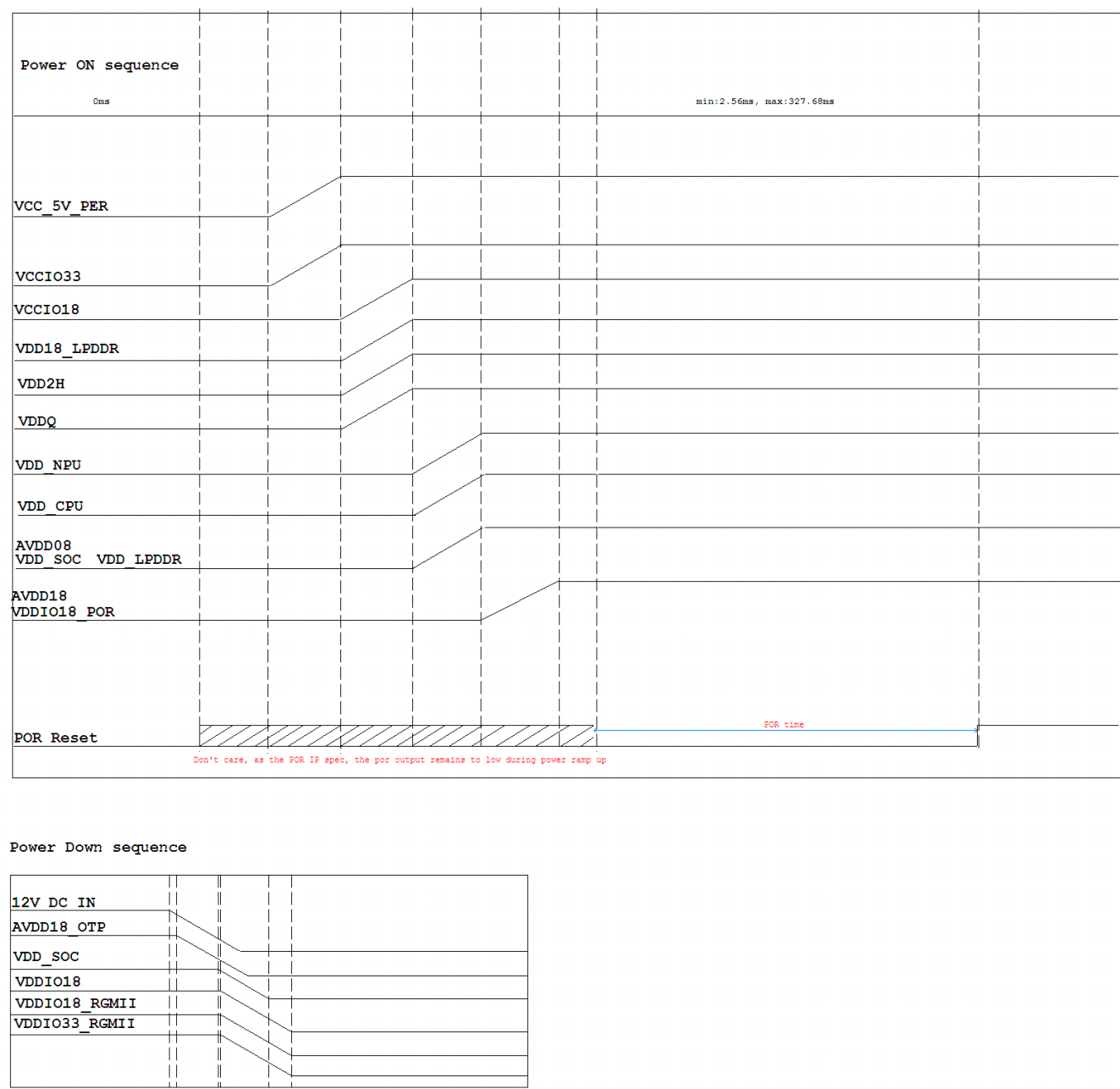


Figure 4: SBC Power-On Sequence

4. Interface and Connector Definitions

This chapter defines the mechanical specifications, pin assignments, electrical connections, and key usage notes for the external interface connectors on the SBC board. These are the core physical characteristics of the SBC as a hardware product.

4.1. 40-Pin IO Expansion Header

This header brings out a rich set of system signals, offering great flexibility for users to expand custom hardware functions (such as connecting sensors, actuators, communication modules, etc.). It includes multiplexed I²C, I²S, SPI, UART, PWM, JTAG, and general-purpose GPIO signals.

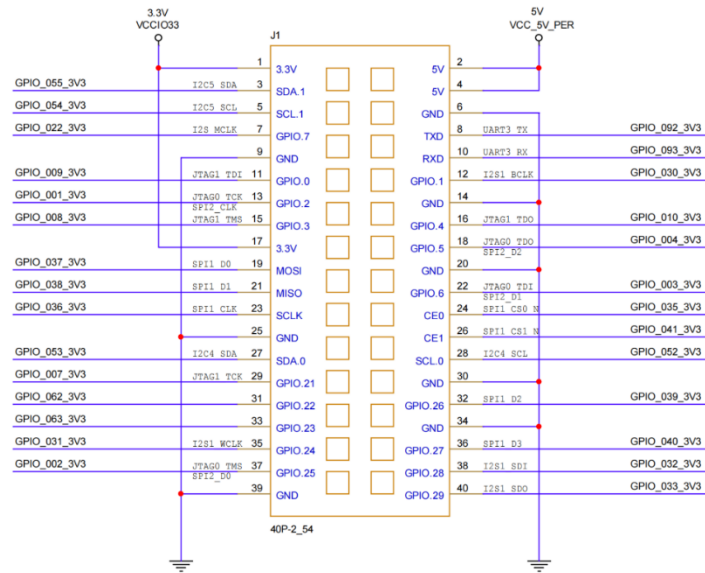


Figure 5: 40-Pin IO Expansion Header

Item	Specification
Connector	J1, 2x20 Pin, 2.54mm pitch, straight pin header (Manufacturer/Model: Lian-sheng PH-00602).
Pin Voltage	All signal pins operate at 3.3V. Do not apply signals exceeding this voltage to avoid damaging the chip.
Power Output	Provides 5.0V and 3.3V power pins for powering external daughter boards or sensors. Pay attention to total power limits.
Key Features	Signals are highly multiplexed. Most pins can be configured via software as general-purpose GPIO or other peripheral functions in addition to their default functions.

Pin Definition Table:

Table 2: SBC 40-Pin Expansion Header Pin Definitions

Pin	Name	Type	Voltage	Default Function	Description
1	3.3V	Power	3.3V		3.3V Power Output
2	5V	Power	5.0V		5V Power Output
3	SDA.1	I/O	3.3V	I2C Data	I2C5_SDA, muxable as GPIO55

Table 2: SBC 40-Pin Expansion Header Pin Definitions

Pin	Name	Type	Voltage	Default Function	Description
4	5V	Power	5.0V		5V Power Output
5	SCL1	Output	3.3V	I2C Clock	I2C5_SCL, muxable as GPIO54
6	GND	Ground			Signal Ground
7	GPIO_022	I/O	3.3V	GPIO	I2S_MCLK, muxable as GPIO22
8	TXD	Output	3.3V	UART TX	UART3_TX, muxable as GPIO92
9	GND	Ground			Signal Ground
10	RXD	Input	3.3V	UART RX	UART3_RX, muxable as GPIO93
11	GPIO_009	I/O	3.3V	GPIO	JTAG1_TDI, muxable as GPIO09
12	GPIO_030	I/O	3.3V	GPIO	I2S1_BCLK, muxable as GPIO30
13	GPIO_001	I/O	3.3V	GPIO	JTAG0_TCK, muxable as SPI2_CLK or GPIO1
14	GND	Ground			Signal Ground
15	GPIO_008	I/O	3.3V	GPIO	JTAG1_TMS, muxable as GPIO8
16	GPIO_010	I/O	3.3V	GPIO	JTAG1_TDO, muxable as GPIO10
17	3.3V	Power	3.3V		3.3V Power Output
18	GPIO_004	I/O	3.3V	GPIO	JTAG0_TDO, muxable as SPI2_D2 or GPIO4
19	MOSI	Output	3.3V	SPI MOSI	SPI1_D0, muxable as GPIO37
20	GND	Ground			Signal Ground
21	MISO	Input	3.3V	SPI MISO	SPI1_D1, muxable as GPIO38
22	GPIO_003	I/O	3.3V	GPIO	JTAG0_TDI, muxable as SPI2_D1 or GPIO3
23	SCLK	Output	3.3V	SPI Clock	SPI1_CLK, muxable as GPIO36
24	CE0	Output	3.3V	SPI Chip Select 0	SPI1_CS0_N, muxable as GPIO35
25	GND	Ground			Signal Ground
26	CE1	Output	3.3V	SPI Chip Select 1	SPI1_CS1_N, muxable as GPIO41 or PWM2
27	SDA.0	I/O	3.3V	I2C Data	I2C4_SDA, muxable as GPIO53
28	SCL.0	Output	3.3V	I2C Clock	I2C4_SCL, muxable as GPIO52
29	GPIO_007	I/O	3.3V	GPIO	JTAG1_TCK, muxable as GPIO7
30	GND	Ground			Signal Ground
31	GPIO_062	I/O	3.3V	GPIO	Dedicated GPIO62
32	GPIO_039	I/O	3.3V	GPIO	SPI1_D2, muxable as GPIO39
33	GPIO_063	I/O	3.3V	GPIO	Dedicated GPIO63
34	GND	Ground			Signal Ground
35	GPIO_031	I/O	3.3V	GPIO	I2S1_WCLK, muxable as GPIO31
36	GPIO_040	I/O	3.3V	GPIO	SPI1_D3, muxable as GPIO40 or PWM1
37	GPIO_002	I/O	3.3V	GPIO	JTAG0_TMS, muxable as SPI2_D0 or GPIO2
38	GPIO_032	I/O	3.3V	GPIO	I2S1_SDI, muxable as GPIO32

Table 2: SBC 40-Pin Expansion Header Pin Definitions

Pin	Name	Type	Voltage	Default Function	Description
39	GND	Ground			Signal Ground
40	GPIO_033	I/O	3.3V	GPIO	I2S1_SDO, muxable as GPIO33

4.2. MIPI CSI Camera Interface (J10)

This interface is a dedicated 4-lane MIPI CSI input for connecting camera modules.

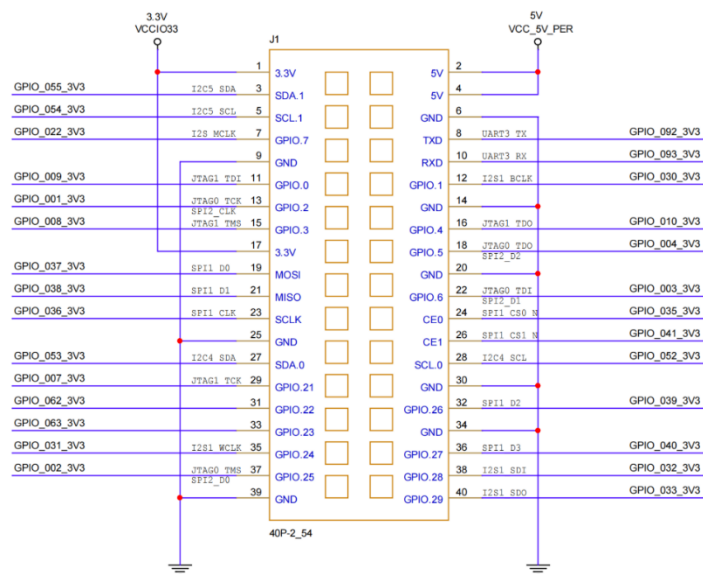


Figure 6: MIPI CSI Interface

Item	Specification
Connector	0.5mm pitch, 22 Pin, slide-in FPC connector (Manufacturer/Model: JS AFC11-S22ICC-00).
Gold Finger Orientation	Note: When connecting the FPC cable, the gold fingers must face toward the onboard RJ45 Ethernet port.
Physical Layer	Interface signals are driven by the CSI0/1 PHY of the EIC7700 SoC, supporting D-PHY and C-PHY modes.
Compatible Cameras	Use MIPI CSI camera modules that match the interface's electrical specifications (e.g., voltage, termination).

About the MIPI PHY Modes of the EIC7700 SoC

The onboard EIC7700 SoC provides a flexible MIPI CSI PHY supporting D-PHY and C-PHY protocols. The SBC hardware design configures it to a specific mode. The following are the SoC chip-level (Ball Level) pin function definitions for advanced hardware integration or in-depth debugging:

Table 3: EIC7700 C-PHY / D-PHY Mode Description

Ball Number	Ball Name	D-PHY Mode	C-PHY Mode
AN27	MIPI_CSI0_CKN	MIPI_CSI0_CKN	MIPI_CSI0_C1
AP27	MIPI_CSI0_CKP	MIPI_CSI0_CKP	MIPI_CSI0_C0

Table 3: EIC7700 C-PHY / D-PHY Mode Description

Ball Number	Ball Name	D-PHY Mode	C-PHY Mode
AN26	MIPI_CSI0_D0N	MIPI_CSI0_D0N	MIPI_CSI0_B0
AP26	MIPI_CSI0_D0P	MIPI_CSI0_D0P	MIPI_CSI0_A0
AN28	MIPI_CSI0_D1N	MIPI_CSI0_D1N	MIPI_CSI0_B1
AP28	MIPI_CSI0_D1P	MIPI_CSI0_D1P	MIPI_CSI0_A1
AP30	MIPI_CSI1_CKN	MIPI_CSI1_CKN	MIPI_CSI1_C1
AN30	MIPI_CSI1_CKP	MIPI_CSI1_CKP	MIPI_CSI1_C0
AN31	MIPI_CSI1_D0N	MIPI_CSI1_D0N	MIPI_CSI1_B0
AP31	MIPI_CSI1_D0P	MIPI_CSI1_D0P	MIPI_CSI1_A0
AN29	MIPI_CSI1_D1N	MIPI_CSI1_D1N	MIPI_CSI1_B1
AP29	MIPI_CSI1_D1P	MIPI_CSI1_D1P	MIPI_CSI1_A1
AN33	MIPI_CSI2_CKN	MIPI_CSI2_CKN	MIPI_CSI2_C1
AP33	MIPI_CSI2_CKP	MIPI_CSI2_CKP	MIPI_CSI2_C0
AN32	MIPI_CSI2_D0N	MIPI_CSI2_D0N	MIPI_CSI2_B0
AP32	MIPI_CSI2_D0P	MIPI_CSI2_D0P	MIPI_CSI2_A0
AM34	MIPI_CSI2_D1N	MIPI_CSI2_D1N	MIPI_CSI2_B1
AM33	MIPI_CSI2_D1P	MIPI_CSI2_D1P	MIPI_CSI2_A1
AK33	MIPI_CSI3_CKN	MIPI_CSI3_CKN	MIPI_CSI3_C1
AK34	MIPI_CSI3_CKP	MIPI_CSI3_CKP	MIPI_CSI3_C0
AJ33	MIPI_CSI3_D0N	MIPI_CSI3_D0N	MIPI_CSI3_B0
AJ34	MIPI_CSI3_D0P	MIPI_CSI3_D0P	MIPI_CSI3_A0
AL33	MIPI_CSI3_D1N	MIPI_CSI3_D1N	MIPI_CSI3_B1
AL34	MIPI_CSI3_D1P	MIPI_CSI3_D1P	MIPI_CSI3_A1
AG33	MIPI_CSI4_CKN	MIPI_CSI4_CKN	MIPI_CSI4_C1
AG34	MIPI_CSI4_CKP	MIPI_CSI4_CKP	MIPI_CSI4_C0
AH33	MIPI_CSI4_D0N	MIPI_CSI4_D0N	MIPI_CSI4_B0
AH34	MIPI_CSI4_D0P	MIPI_CSI4_D0P	MIPI_CSI4_A0
AF33	MIPI_CSI4_D1N	MIPI_CSI4_D1N	MIPI_CSI4_B1
AF34	MIPI_CSI4_D1P	MIPI_CSI4_D1P	MIPI_CSI4_A1
AF30	MIPI_CSI5_CKN	MIPI_CSI5_CKN	MIPI_CSI5_C1
AF31	MIPI_CSI5_CKP	MIPI_CSI5_CKP	MIPI_CSI5_C0
AE30	MIPI_CSI5_D0N	MIPI_CSI5_D0N	MIPI_CSI5_B0
AE31	MIPI_CSI5_D0P	MIPI_CSI5_D0P	MIPI_CSI5_A0
AG30	MIPI_CSI5_D1N	MIPI_CSI5_D1N	MIPI_CSI5_B1
AG31	MIPI_CSI5_D1P	MIPI_CSI5_D1P	MIPI_CSI5_A1

4.2.1. C-PHY Mode

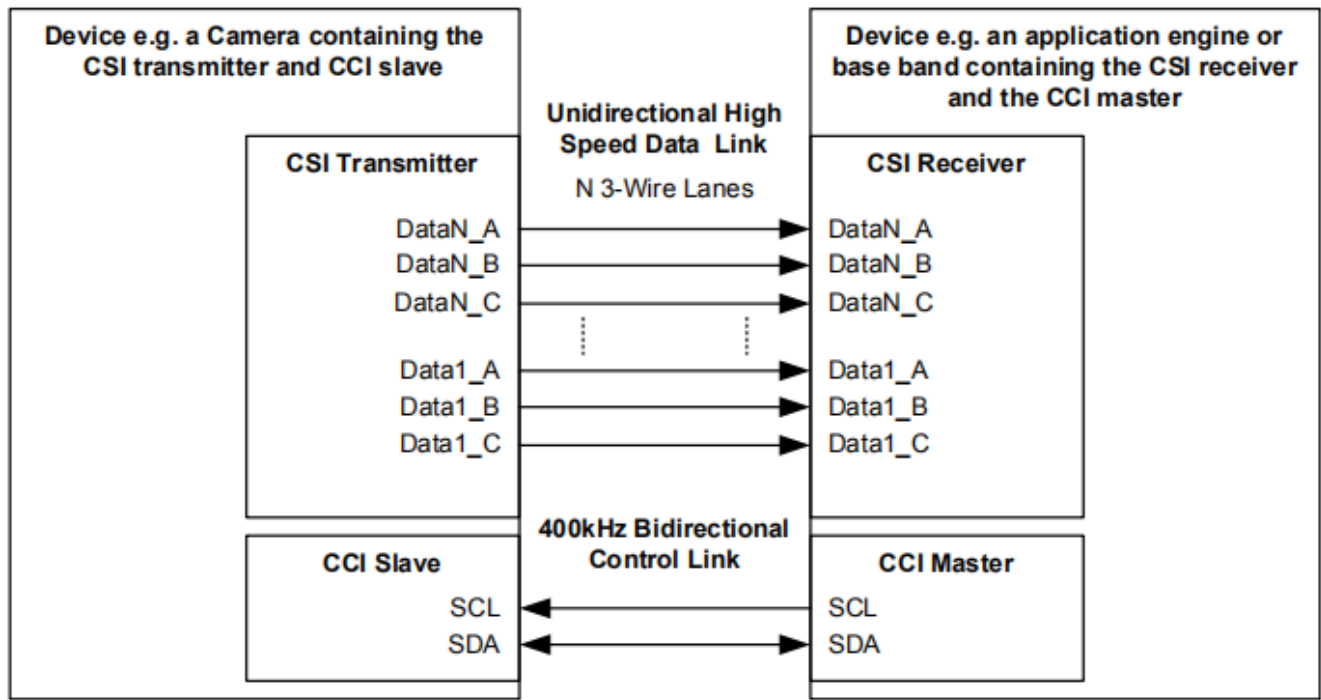


Figure 7: C-PHY Connection Diagram

4.2.2. D-PHY Mode

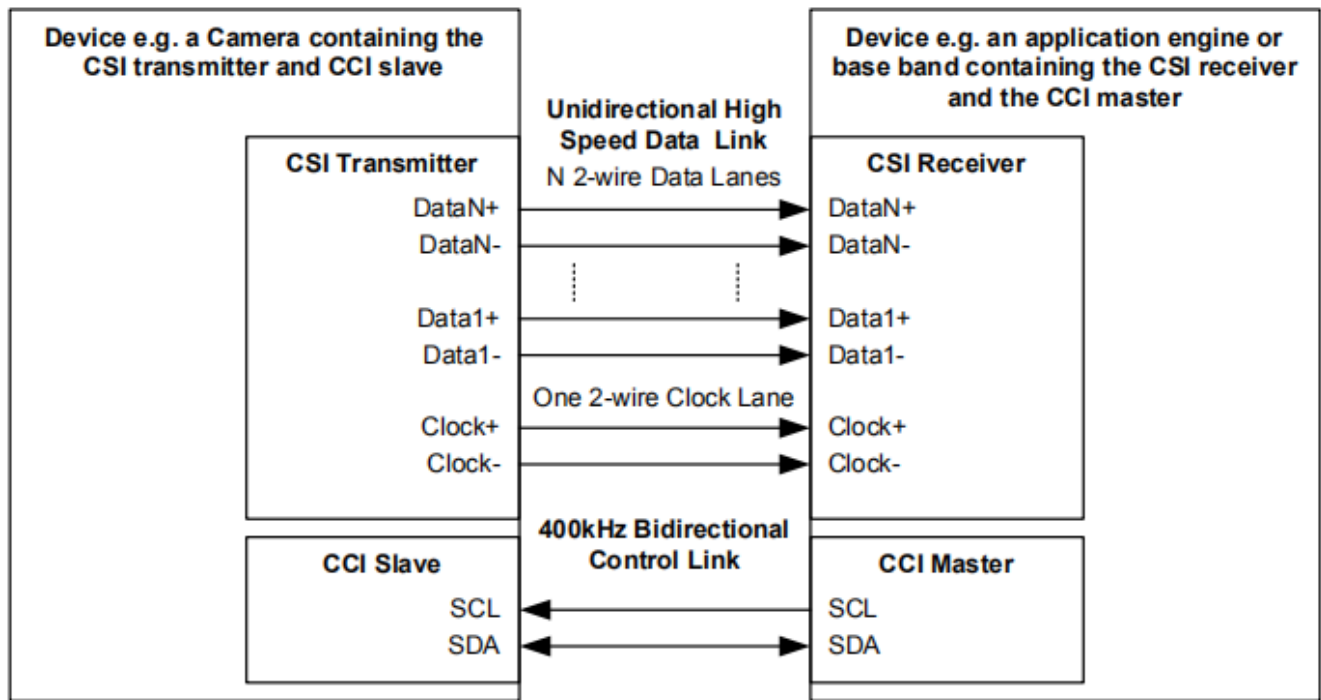


Figure 8: D-PHY Connection Diagram

The MIPI CSI D-PHY of the EIC7700 SoC supports flexible lane combination configurations. Every two adjacent 2-Lane PHYs can be combined into a single 4-Lane PHY.

Specifically: CSI0 and CSI1 can be combined, CSI2 and CSI3 can be combined, and CSI4 and CSI5 can be combined. When two PHYs are combined into 4-Lane mode, only the clock lane of the lower-numbered PHY (primary PHY) in the pair is used, while the clock lane of the higher-numbered PHY (secondary PHY) is unused. The data lanes are remapped such that the two data lanes (Lane0, Lane1) of the secondary PHY serve as the additional data lanes (Lane2, Lane3) of the primary PHY.

The following table uses the combination of CSI0 and CSI1 as an example, detailing the signal function mapping of the corresponding chip balls in 2-Lane independent mode and 4-Lane combined mode.

Table 4: EIC7700 D-PHY 2/4 Lane Mode

Ball Number	Ball Name	D-PHY 2 Lane Mode	D-PHY 4 Lane Mode
AN27	MIPI_CSI0_CKN	MIPI_CSI0_CKN	MIPI_CSI0_CKN
AP27	MIPI_CSI0_CKP	MIPI_CSI0_CKP	MIPI_CSI0_CKP
AN26	MIPI_CSI0_D0N	MIPI_CSI0_D0N	MIPI_CSI0_D0N
AP26	MIPI_CSI0_D0P	MIPI_CSI0_D0P	MIPI_CSI0_D0P
AN28	MIPI_CSI0_D1N	MIPI_CSI0_D1N	MIPI_CSI0_D1N
AP28	MIPI_CSI0_D1P	MIPI_CSI0_D1P	MIPI_CSI0_D1P
AP30	MIPI_CSI1_CKN	MIPI_CSI1_CKN	-
AN30	MIPI_CSI1_CKP	MIPI_CSI1_CKP	-
AN31	MIPI_CSI1_D0N	MIPI_CSI1_D0N	MIPI_CSI0_D2N
AP31	MIPI_CSI1_D0P	MIPI_CSI1_D0P	MIPI_CSI0_D2P
AN29	MIPI_CSI1_D1N	MIPI_CSI1_D1N	MIPI_CSI0_D3N
AP29	MIPI_CSI1_D1P	MIPI_CSI1_D1P	MIPI_CSI0_D3P

4.3. MIPI DSI/CSI Multiplexed Interface (J9)

This interface is a hardware multiplexed interface that can be configured via software as either a MIPI DSI display output or a MIPI CSI (CSI23) camera input. This interface shares the display layer with HDMI and cannot be used simultaneously.

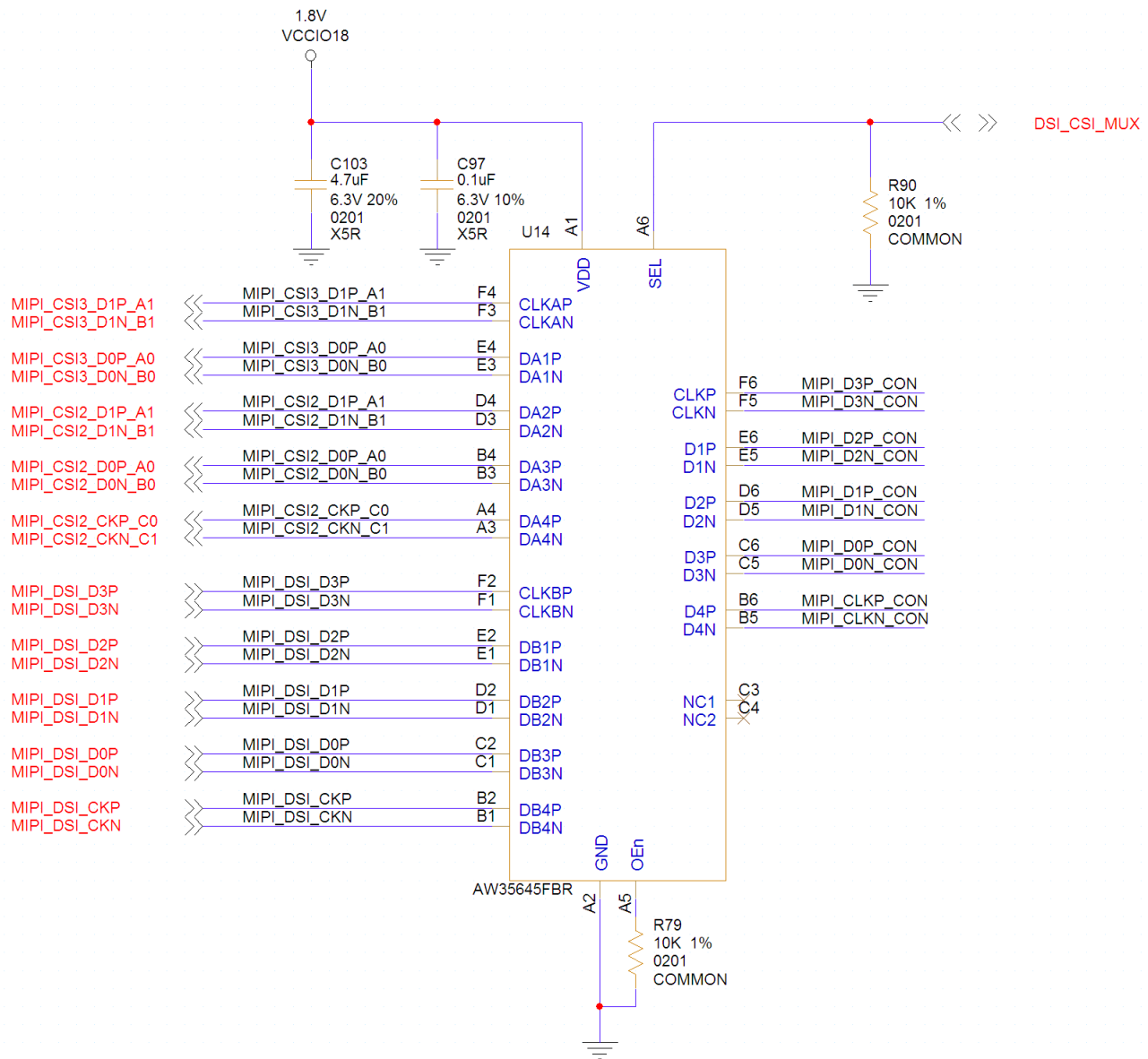


Figure 9: MIPI DSI/CSI Interface Diagram

Item	Specification
Connector	0.5mm pitch, 22 Pin, slide-in FPC connector (same model as J10).
Gold Finger Orientation	Note: When connecting the FPC cable, the gold fingers must face toward the onboard RJ45 Ethernet port.
Mode Switching	Controlled by GPIO_028 signal level: High = DSI mode, Low = CSI mode. Default state is determined by software.
Multiplexing Warning	Physically shares the display processing unit with HDMI; cannot output video simultaneously with the HDMI interface.
Configuration Method	Switch modes by loading different Device Tree Blob (DTB) files in the operating system. See the user guide for details.

DSI Mode Pin Definitions:

When the interface is configured as a MIPI DSI display output, the pin definitions are as follows:

Table 5: MIPI DSI Interface Pin Definitions

Signal Name	Description	Type	Power Rail
MIPI_DSI_CKN	MIPI DSI Clock Lane Negative	Output	VPH_DSI
MIPI_DSI_CKP	MIPI DSI Clock Lane Positive	Output	VPH_DSI
MIPI_DSI_D0N	MIPI DSI Data Lane 0 Negative	Output	VPH_DSI
MIPI_DSI_D0P	MIPI DSI Data Lane 0 Positive	Output	VPH_DSI
MIPI_DSI_D1N	MIPI DSI Data Lane 1 Negative	Output	VPH_DSI
MIPI_DSI_D1P	MIPI DSI Data Lane 1 Positive	Output	VPH_DSI
MIPI_DSI_D2N	MIPI DSI Data Lane 2 Negative	Output	VPH_DSI
MIPI_DSI_D2P	MIPI DSI Data Lane 2 Positive	Output	VPH_DSI
MIPI_DSI_D3N	MIPI DSI Data Lane 3 Negative	Output	VPH_DSI
MIPI_DSI_D3P	MIPI DSI Data Lane 3 Positive	Output	VPH_DSI

Warning

When this interface is configured in CSI mode, its pin definitions are related to the SoC' s CSI2/3 PHY. Refer to the schematic for detailed connections.

4.4. PCIe Gen3 x4 Expansion Interface (J6)

Used for connecting high-speed expansion devices such as M.2 SSDs, AI accelerator cards, network cards, etc.

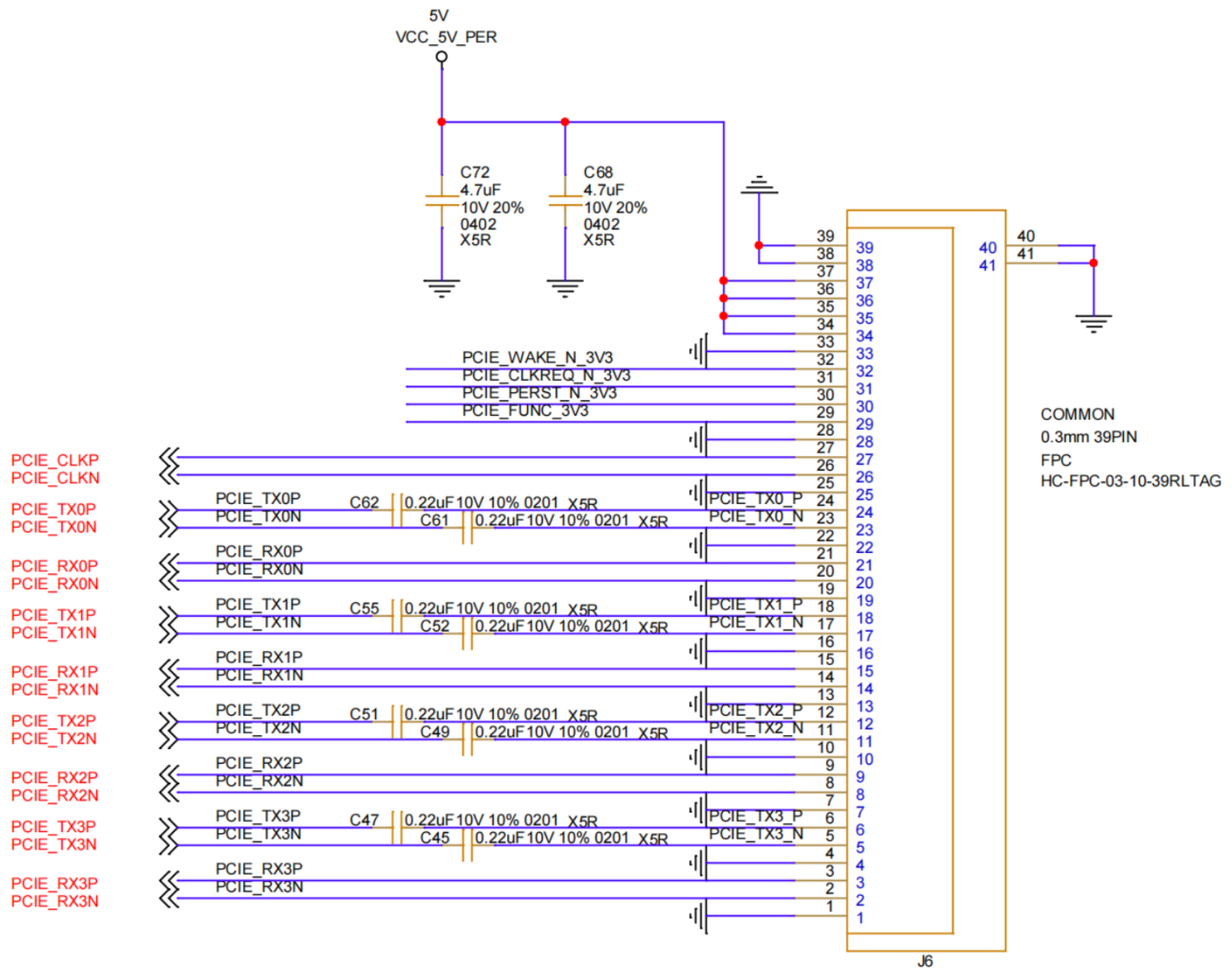


Figure 10: PCIe Interface Diagram

Item	Specification
Connector	0.3mm pitch, 39 Pin, flip-lid FPC connector (Manufacturer/Model: HCTL HC-FPC-03-10-39RLTAG).
Gold Finger Orientation	Note: When connecting the FPC cable, the gold fingers must face downward (toward the PCB board).
Protocol	PCI Express Gen3 x4.
Operating Modes	Supports Root Complex (RC, host) and Endpoint (EP, device) modes, determined by hardware design or software configuration.

Warning

- External Clock:** Regardless of RC or EP mode, a 100MHz HCSL differential reference clock must be provided by the connected device (host board or daughter card) to the SBC's PCIE_CLKP/N pins.
- Power Connection:**
 - When the SBC operates as RC, the 5V pins on this interface can supply power to the

daughter card.

- When the SBC operates as **EP**, do NOT connect the 5V power pins on this interface. The SBC must be powered by the external host board through other means.

Pin Definitions:

Table 6: PCIe Interface Pin Definitions

Signal Name	Description	Type	Power Rail
D0_PCIE_CLKN	PCIe Reference Clock Negative	Input	D0_VP_PCIE
D0_PCIE_CLKP	PCIe Reference Clock Positive	Input	D0_VP_PCIE
D0_PCIE_RX0N	PCIe Receive Lane 0 Negative	Input	D0_VP_PCIE
D0_PCIE_RX0P	PCIe Receive Lane 0 Positive	Input	D0_VP_PCIE
D0_PCIE_RX1N	PCIe Receive Lane 1 Negative	Input	D0_VP_PCIE
D0_PCIE_RX1P	PCIe Receive Lane 1 Positive	Input	D0_VP_PCIE
D0_PCIE_RX2N	PCIe Receive Lane 2 Negative	Input	D0_VP_PCIE
D0_PCIE_RX2P	PCIe Receive Lane 2 Positive	Input	D0_VP_PCIE
D0_PCIE_RX3N	PCIe Receive Lane 3 Negative	Input	D0_VP_PCIE
D0_PCIE_RX3P	PCIe Receive Lane 3 Positive	Input	D0_VP_PCIE
D0_PCIE_TX0N	PCIe Transmit Lane 0 Negative	Output	D0_VP_PCIE
D0_PCIE_TX0P	PCIe Transmit Lane 0 Positive	Output	D0_VP_PCIE
D0_PCIE_TX1N	PCIe Transmit Lane 1 Negative	Output	D0_VP_PCIE
D0_PCIE_TX1P	PCIe Transmit Lane 1 Positive	Output	D0_VP_PCIE
D0_PCIE_TX2N	PCIe Transmit Lane 2 Negative	Output	D0_VP_PCIE
D0_PCIE_TX2P	PCIe Transmit Lane 2 Positive	Output	D0_VP_PCIE
D0_PCIE_TX3N	PCIe Transmit Lane 3 Negative	Output	D0_VP_PCIE
D0_PCIE_TX3P	PCIe Transmit Lane 3 Positive	Output	D0_VP_PCIE
D0_PCIE_CLKREQ_N	PCIe Clock Request	Output	VDDIO18
D0_PCIE_PERST_N	PCIe Reset	I/O	VDDIO18
D0_PCIE_WAKE_N	PCIe Link Wake	I/O	VDDIO18

4.5. Fan Interface (J2)

Used to connect a 4-wire PWM-controlled fan for enhanced system cooling.

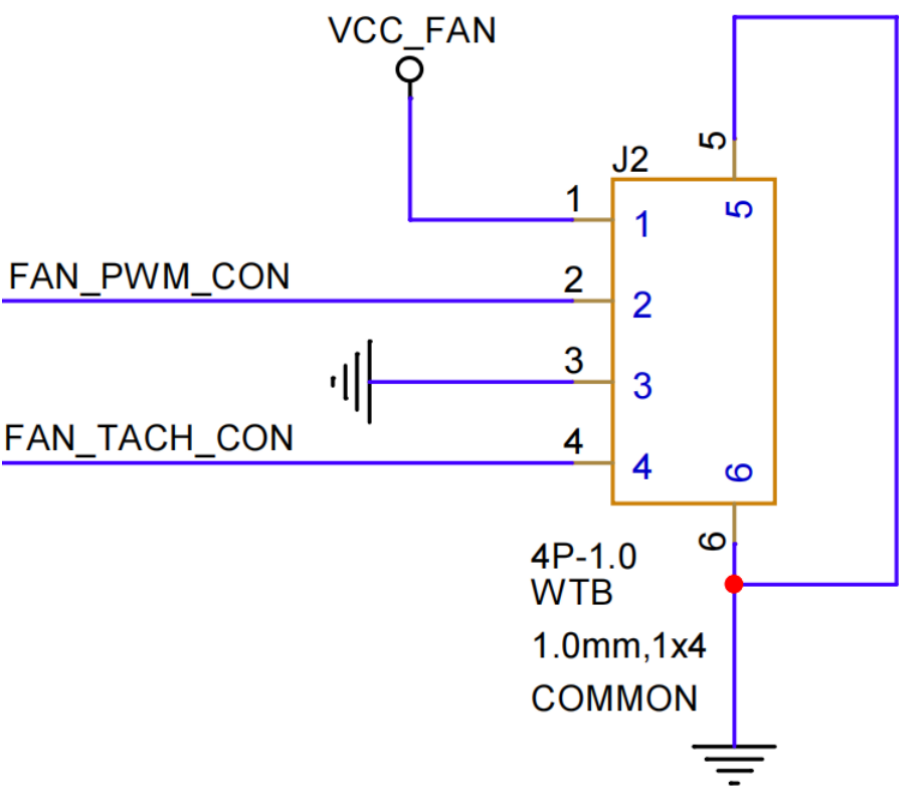


Figure 11: Fan Interface Diagram

Item	Specification
Connector	1.0mm pitch, 4 Pin, SMD connector (Manufacturer/Model: SHOUHAN 1.0-4P LT).
Power	Provides 5V power, compatible with common 5V fans.
Function	Supports PWM speed control and tachometer signal feedback for monitoring fan status in the system.

Pin Definitions:

Pin	Signal Name	Description	Type
1	5.0V	Fan Power (5V)	Power
2	PWM	Fan Speed PWM Control Signal	Output
3	GND	Power Ground	Ground
4	TACH	Fan Speed Tachometer Feedback	Input

4.6. RTC Battery Interface (J12)

Used to connect a 3V coin cell battery (e.g., CR2032) to power the onboard real-time clock (RTC) circuit and maintain time and date when the main power is disconnected.

5. Mechanical Dimensions

The precise mechanical dimensions and mounting hole locations of the SBC are shown in the figure below. All dimensions are in millimeters (mm).

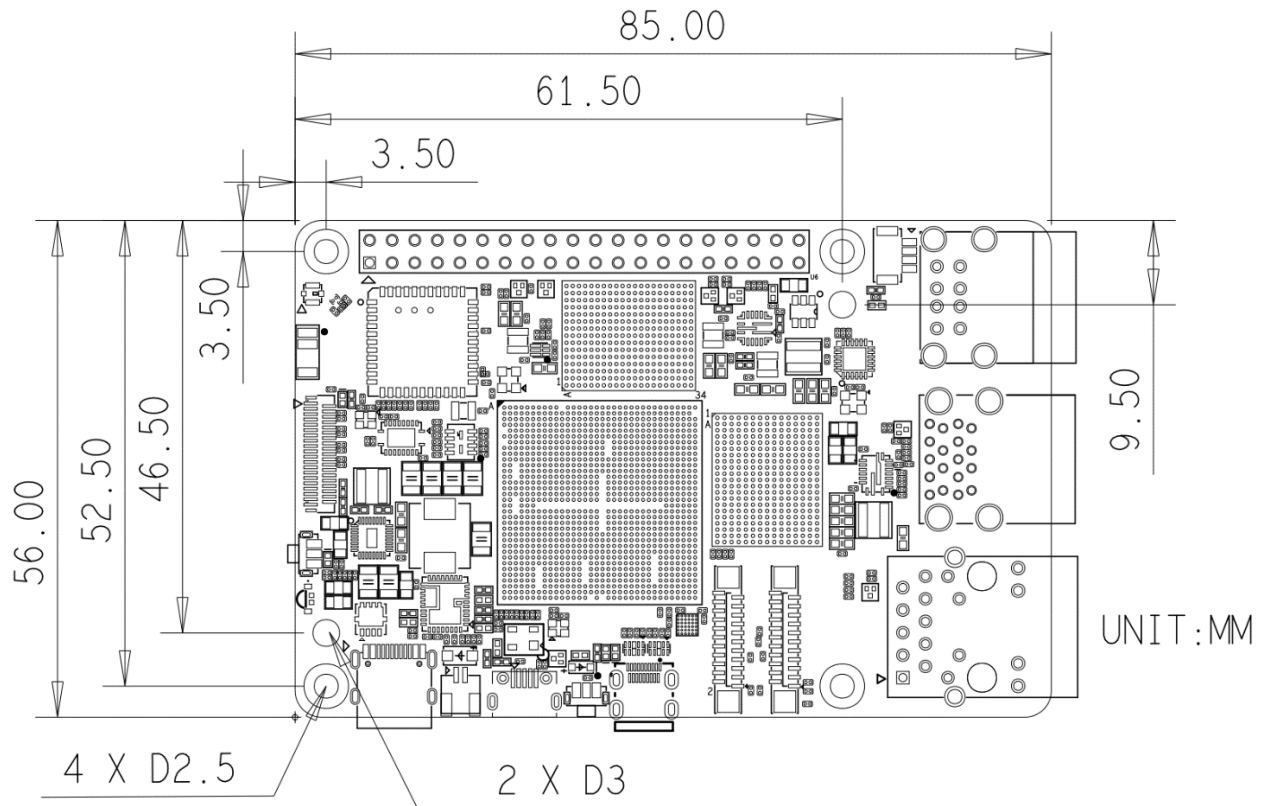


Figure 13: Mechanical Dimensions